Dquez Carr
MCY 650
Lab 2 Network Packet Capture
September, 2024

Lab 2. Network Packet Capture

Instructions:

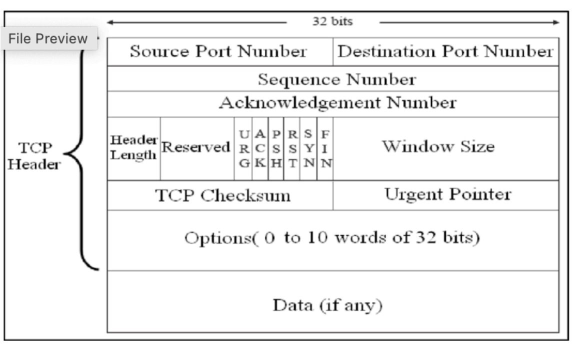
- You will work on this lab only using your Kali Linux virtual machine only. Please see virtual machine access instructions on Canvas.
- Finish the lab by following the lab instructions step by step.
- Answer the questions at the end of the lab. Questions are highlighted yellow. Each question is worth 50 points. Please embed your answer right after each question in this document. This updated document will be your lab report. Please submit it on Canvas.

1. Introduction:

In this lab, we are going to explore network packet capturing using Wireshark. TCP connections start with a TCP 3-way handshake. The first 3 TCP packets related to a TCP connection are the SYN packet, SYN/ACK packet, and ACK packets. The process is as follows:

SYN: Client to Server, SYN/ACK: Server to Client, ACK: Client to Server

What is SYN, SYN/ACK, and ACK anyway? SYN and ACK are TCP flags in the TCP header. The following diagram shows the TCP header with control flags. You can see there are other control flags also: URG, PSH, RST, FIN. So the SYN packet means the SYN bit is set as 1. Similarly, SYN/ACK packet means both SYN bit and ACK bit are set as 1. Other TCP flags (URG, PSH, RST, FIN) not used during TPC 3-way handshake are set as zeros.



2. Using Wireshark:

If you login as root, you can type **wireshark** in the terminal window; otherwise, you need to prepend it with sudo to run it. Next, click on the name of your network interface (it is likely to be something like eth0 for ethernet network card). Wireshark will then start capturing network packets.

Leave Wireshark running, then launch a browser from Kali, say, Firefox. (You can do this by clicking on Applications -> Firefox ESR).

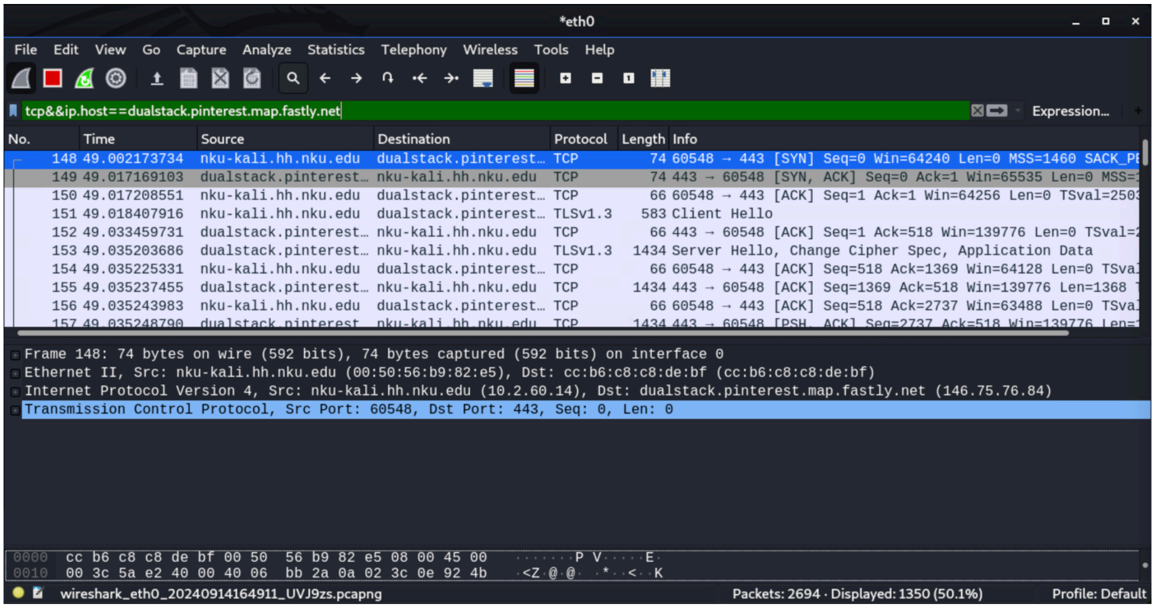
Go back to the Wireshark window. Click View -> Name Resolution on the menu bar, then make sure to check the checkbox next to “Resolve Network Addresses”.

Go back to the Firefox web browser and visit www.google.com. Repeat this multiple times if you wish to generate more packets.

In the display filter field of Wireshark, type <tcp && ip.host == www.google.com>. You will see packets related to Google. Because the SYN packet is the first packet related to the TCP three-way handshake, you can sort it by clicking on the “No.” column because the “No.” column orders packets based on their capture time.

You can see from the above screen capture that there are other packets listed after the 3-way handshake. What if we only want to see the SYN packets? Type the following in the filter field: <tcp && ip.host==www.google && tcp.flags.syn==1>. If you close and reopen your connection to google.com, you will increase the number of SYN packets seen here.

Please connect to another website of your choice by browser and use Wireshark to capture the SYN, SYN/ACK, and ACK packets. Show your screenshot of the packets captured.



Next, we will learn how to examine packets related to the ping command. Run the following command in the terminal: ping www.google.com. Then go back to the Wireshark window and enter <icpm && ip.host==www.google.com> in the filter field, you will see Echo request and Echo reply packets generated by ping command.

Please ping another website of your choice at the command line and use Wireshark to capture the ICMP packets generated by the ping command. Show your screenshot of the packets captured.